

RAFFLES INSTITUTION  
2012 YEAR 6 PRELIMINARY EXAMINATION

Higher 1



# CHEMISTRY

8872/01

Paper 1 Multiple Choice

21 September 2012

50 minutes

Additional Materials:      Multiple Choice Answer Sheet  
                                         Data Booklet

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## READ THESE INSTRUCTIONS FIRST

**Do not open this question booklet until you are told to do so.**

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number in the spaces provided on the Answer Sheet.

There are **thirty** questions on this paper. Answer **all** questions.

For each question there are four possible answers **A**, **B**, **C** and **D**.

Choose the **one** you consider to be correct and record your choice with a soft pencil on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

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This document consists of **11** printed pages.

## Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 Sulfur dichloride dioxide,  $\text{SO}_2\text{Cl}_2$ , reacts with water to give a mixture of sulfuric acid and hydrochloric acid. How many moles of sodium hydroxide,  $\text{NaOH}$ , would be needed to neutralise the solution formed by adding one mole of  $\text{SO}_2\text{Cl}_2$  to an excess of water?

**A** 1                      **B** 2                      **C** 3                      **D** 4

- 2 At one stage in the radioactive decay of the isotope  $^{235}_{92}\text{U}$ , the isotope  $^{211}_{82}\text{Pb}$  is present. How many  $\alpha$  particles and  $\beta$  particles will be emitted in the decay?

An  $\alpha$  particle is  $^4_2\text{He}^{2+}$  and a  $\beta$  particle is  $^0_{-1}\text{e}^-$ .

- A** 2  $\alpha$  particles and 6  $\beta$  particles  
**B** 6  $\alpha$  particles and 2  $\beta$  particles  
**C** 6  $\alpha$  particles and 6  $\beta$  particles  
**D** 12  $\alpha$  particles and 12  $\beta$  particles

- 3 *Use of the Data Booklet is relevant to this question.*

The ion  $\text{X}^{4+}$  contains 31 electrons and 46 neutrons. Which statement about  $\text{X}^{4+}$  or  $\text{X}$  is correct?

- A**  $\text{X}^{4+}$  has the electronic configuration  $[\text{Ar}] 3\text{d}^9$ .  
**B**  $\text{X}^{4+}$  is deflected approximately to twice the extent as  $^{23}\text{Na}^+$  when subjected to an electric field.  
**C**  $\text{X}$  is a good conductor of electricity.  
**D** The elemental form of  $\text{X}$  can act as an electrophile.

- 4 Which of the species has a bond angle of  $120^\circ$ ?

- A**  $\text{ICl}_3$   
**B**  $\text{PBr}_3$   
**C**  $\text{AlCl}_3$   
**D**  $\text{NH}_3$

- 5 Ethanol is more soluble in water than is ethyl ethanoate. Which of the following statements correctly accounts for this?

- A** Ethanol is a polar molecule but ethyl ethanoate is non-polar.  
**B** Ethanol is able to dissociate into  $\text{C}_2\text{H}_5^+$  and  $\text{OH}^-$  ions that are readily solvated but ethyl ethanoate is not able to dissociate into ions.  
**C** A hydrogen bond forms between the hydrogen of a water molecule and the oxygen of ethanol.  
**D** A hydrogen bond forms between the hydrogen of the  $-\text{OH}$  group in ethanol and the hydrogen of a water molecule.

- 6 Which row shows the correct number of  $\sigma$  and  $\pi$  bonds present in the respective molecules?

|   | $\text{NH}_3$   | $\text{C}_2\text{H}_4$ | $\text{N}_2$    |
|---|-----------------|------------------------|-----------------|
| A | $4\sigma, 0\pi$ | $1\sigma, 5\pi$        | $3\sigma, 0\pi$ |
| B | $3\sigma, 0\pi$ | $2\sigma, 4\pi$        | $2\sigma, 1\pi$ |
| C | $3\sigma, 0\pi$ | $5\sigma, 1\pi$        | $1\sigma, 2\pi$ |
| D | $3\sigma, 1\pi$ | $6\sigma, 0\pi$        | $0\sigma, 3\pi$ |

- 7 The enthalpy changes of formation of gaseous ethene and ethane at 298 K are  $+52 \text{ kJ mol}^{-1}$  and  $-85 \text{ kJ mol}^{-1}$  respectively.

Consider the following reaction:  $\text{C}_2\text{H}_4(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$

Which statement is true?

- A The C—C bond in ethane has a larger bond energy than the C—C bond in ethene.  
 B The enthalpy change of the reaction is  $-137 \text{ kJ mol}^{-1}$ .  
 C If ethene is converted to ethane via four stages instead of one as shown above, the enthalpy change of the reaction will be less exothermic.  
 D The addition of a catalyst will cause the enthalpy change of reaction to be more exothermic.
- 8 In the commercial electrolysis of brine, a concentrated aqueous solution of sodium chloride, the products are chlorine, hydrogen and sodium hydroxide.

The oxidation reaction is written as:  $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$   
 while the reduction reaction is:  $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$

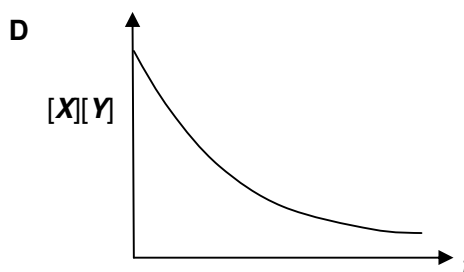
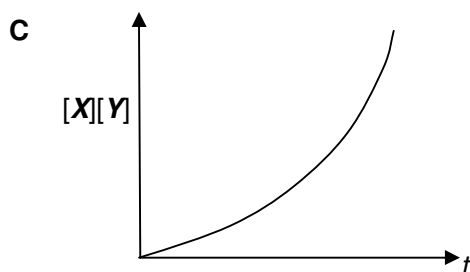
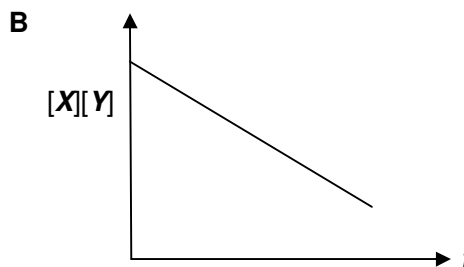
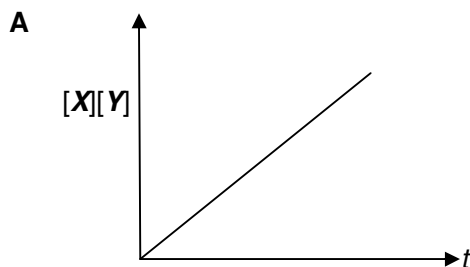
What is the maximum yield of each of these products when brine containing 58.5 kg of sodium chloride is electrolysed?

|   | <i>chlorine</i> | <i>hydrogen</i> | <i>sodium hydroxide</i> |
|---|-----------------|-----------------|-------------------------|
| A | 35.5 kg         | 1 kg            | 40 kg                   |
| B | 35.5 kg         | 2 kg            | 40 kg                   |
| C | 71 kg           | 1 kg            | 40 kg                   |
| D | 71 kg           | 1 kg            | 80 kg                   |

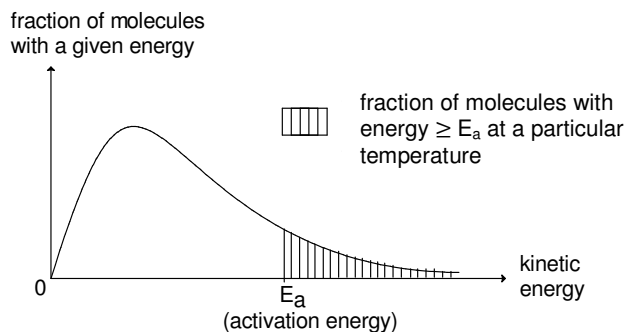
- 9 When  $[X][Y]$ , the product of concentrations of  $X$  and  $Y$ , is plotted against time,  $t$ , for the second-order reaction



which of the following graphs would be obtained?



- 10 For the Maxwell-Boltzmann distribution curve shown below for a chemical reaction in living systems, which statement is **not** true?



- A** The total area under the curve is 1.  
**B** The activation energy will decrease with an increase in temperature.  
**C** When temperature is increased, the peak of the curve becomes lower.  
**D** The shaded area of the curve will decrease when enzymes are denatured.

- 11 The values of the equilibrium constant,  $K_c$ , for the reaction



are  $0.313 \text{ mol dm}^{-3}$  and  $115 \text{ mol dm}^{-3}$  at  $25^\circ\text{C}$  and  $727^\circ\text{C}$  respectively.

What deduction can be made from the information given above?

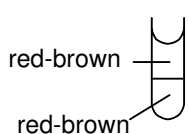
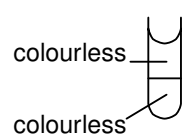
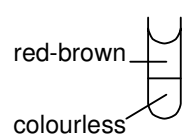
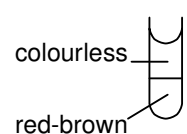
- A** The yield of  $\text{CO}_2$  can be increased by adding some  $\text{Ag}_2\text{CO}_3$  to a system already in equilibrium.  
**B** The activation energy of the reverse reaction is higher than that of the forward reaction.  
**C** Less  $\text{CO}_2$  is formed at equilibrium if the reaction is allowed to take place in another container of a smaller volume.  
**D** More  $\text{CO}_2$  is formed at equilibrium if a suitable catalyst is used.
- 12 The oxide of **G** and the chloride of **H** are separately mixed with water. The two resulting mixtures have the same effect on litmus. Which of the following represents the correct elements?

|          | <b>G</b> | <b>H</b> |
|----------|----------|----------|
| <b>A</b> | Al       | Si       |
| <b>B</b> | Mg       | P        |
| <b>C</b> | Si       | Al       |
| <b>D</b> | P        | Mg       |

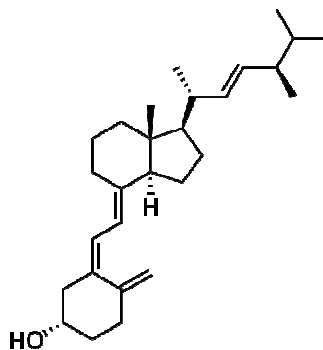
- 13 Which property of Group II elements (from magnesium to barium) or their compounds increases with increasing proton number?

- A** the magnitude of the enthalpy change of hydration of the metal ions  
**B** the magnitude of the lattice energies of the nitrates  
**C** the melting points of the elements  
**D** the ease of losing an electron from an atom of the element

- 14 Aqueous bromine is added to an equal volume of ethene and the mixture shaken together. Which of the following observations would be made after a few minutes?

|                                                                                     |                                                                                     |                                                                                      |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>A</b>                                                                            | <b>B</b>                                                                            | <b>C</b>                                                                             | <b>D</b>                                                                              |
|  |  |  |  |

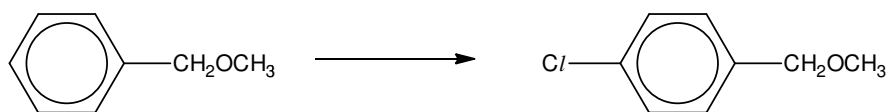
- 15 Vitamin D has the structure



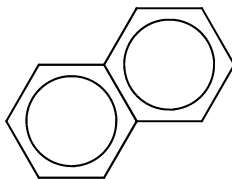
vitamin D

How many possible geometrical isomers exist for vitamin D?

- A**  $2^1$                       **B**  $2^2$                       **C**  $2^3$                       **D**  $2^4$
- 16 Which set of conditions will bring about the transformation shown below?



- A**  $\text{Cl}_2(\text{g})$  in the presence of uv light  
**B**  $\text{Cl}_2(\text{g})$  in the presence of  $\text{AlCl}_3(\text{aq})$   
**C**  $\text{Cl}_2(\text{g})$  in the presence of  $\text{Fe}(\text{s})$   
**D** concentrated  $\text{HCl}(\text{aq})$  with heat
- 17 Naphthalene,  $\text{C}_{10}\text{H}_8$ , is an aromatic compound with the following structure.



It has similar reactions and reactivity as benzene.

Which of the following statements about naphthalene and its derivative is correct?

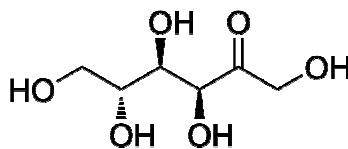
- A** Naphthalene has 12  $\pi$  electrons.  
**B** Naphthalene is a planar molecule.  
**C** Naphthalene decolorises bromine in the presence of uv light.  
**D** There are a total of four aromatic isomers of molecular formula  $\text{C}_{10}\text{H}_6\text{Cl}_2$  with the naphthalene ring structure.

- 18 Four compounds were reacted with reagents containing deuterium, an isotope of hydrogen. [D =  $^2\text{H}$ ]

Which of the following compounds gave a carbon-containing product which does **not** incorporate deuterium?

- A  $\text{CH}_3\text{CH}_2\text{CN}$  with warm NaOD in  $\text{D}_2\text{O}$   
 B  $\text{CH}_3\text{COOCH}_3$  with hot NaOD in  $\text{D}_2\text{O}$   
 C  $\text{CH}_3\text{COCH}_3$  with LiAl/D<sub>4</sub> in dry ether  
 D  $\text{CH}_3\text{CHO}$  with DCN and trace amounts of NaCN

- 19 Fructose is a sweet, white, odourless crystalline solid that can be extracted from many plants.



fructose

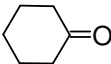
Which statement about fructose is correct?

- A It can undergo oxidation with hot acidified  $\text{K}_2\text{Cr}_2\text{O}_7$ .  
 B It can undergo nucleophilic substitution with  $\text{Br}_2(\text{aq})$ .  
 C 1 mol of fructose reacts with 5 mol of NaOH.  
 D 1 mol of fructose reacts with 1 mol of  $\text{Na}_2\text{CO}_3$ .
- 20 Starting with  $\text{C}_6\text{H}_5\text{COCH}_3$ , which two-stage process does **not** involve a condensation reaction?
- |   | Step 1                           | Step 2                                                                     |
|---|----------------------------------|----------------------------------------------------------------------------|
| A | $\text{NaBH}_4$                  | $\text{CH}_3\text{COC}/$                                                   |
| B | $\text{H}_2\text{N}-\text{NH}_2$ | NaOH                                                                       |
| C | $\text{I}_2$ with NaOH           | $\text{C}_2\text{H}_5\text{OH}/\text{concentrated } \text{H}_2\text{SO}_4$ |
| D | HCN with trace NaCN              | NaOH                                                                       |
- 21 Which reagent could be used to distinguish between  $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CHO}$  and  $\text{CH}_3\text{COCH}_2\text{CH}_2\text{OH}$ ?

- A acidified potassium dichromate(VI)  
 B phosphorus pentachloride  
 C 2,4-dinitrophenylhydrazine  
 D Fehling's reagent

- 22 Compounds **X**, **Y** and **Z** all react with 2,4-dinitrophenylhydrazine but only two of them will cause a reduction in the oxidation number of the metal present in Tollens' reagent.

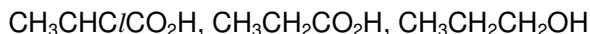
Which combination could be **X**, **Y** and **Z**?

|          | <b>X</b>                           | <b>Y</b>                         | <b>Z</b>                                                                            |
|----------|------------------------------------|----------------------------------|-------------------------------------------------------------------------------------|
| <b>A</b> | $\text{CH}_3\text{CH}_2\text{CHO}$ | $\text{C}_6\text{H}_5\text{CHO}$ |  |
| <b>B</b> | $\text{CH}_3\text{CH}_2\text{CHO}$ | $\text{C}_6\text{H}_5\text{CHO}$ | $\text{CH}_3\text{CH}_2\text{COOCH}_3$                                              |
| <b>C</b> | $\text{CH}_3\text{CONH}_2$         | $\text{CH}_3\text{COCH}_3$       | $\text{C}_6\text{H}_5\text{CHO}$                                                    |
| <b>D</b> | $\text{CH}_3\text{CHO}$            | $\text{C}_6\text{H}_5\text{CHO}$ | $\text{CH}_3\text{CH}_2\text{CHO}$                                                  |

- 23 A compound, **Q**, was boiled with aqueous sodium hydroxide and the resulting mixture was cooled and acidified with dilute sulfuric acid. The final products included a compound  $\text{C}_3\text{H}_6\text{O}_2$  and an alcohol which gave a positive triiodomethane test. Which of the following formulae could represent **Q**?

- A**  $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$   
**B**  $\text{CH}_3\text{CH}_2\text{OCOCH}_3$   
**C**  $\text{CH}_2(\text{OH})\text{CH}_2\text{CH}_2\text{COCH}_3$   
**D**  $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$

- 24 0.1 mol of each of the three compounds below were added separately to 1 dm<sup>3</sup> of water.



Which of the following shows the order of increasing pH of the solutions formed?

- Increasing pH  
 $\xrightarrow{\hspace{10em}}$
- A**  $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}, \text{CH}_3\text{CHC}/\text{CO}_2\text{H}, \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$   
**B**  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}, \text{CH}_3\text{CHC}/\text{CO}_2\text{H}, \text{CH}_3\text{CH}_2\text{CO}_2\text{H}$   
**C**  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}, \text{CH}_3\text{CH}_2\text{CO}_2\text{H}, \text{CH}_3\text{CHC}/\text{CO}_2\text{H}$   
**D**  $\text{CH}_3\text{CHC}/\text{CO}_2\text{H}, \text{CH}_3\text{CH}_2\text{CO}_2\text{H}, \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

- 25 In green chemistry, the efficiency of a chemical reaction can be measured by atom economy, where

$$\% \text{ atom economy} = \frac{\text{Molecular weight of desired product}}{\text{Molecular weight of all reactants}} \times 100$$

Which of the following method of producing ethanol results in the greatest % atom economy?

- A**  $\text{C}_2\text{H}_5\text{Br} + \text{NaOH} \rightarrow \text{C}_2\text{H}_5\text{OH} + \text{NaBr}$   
**B**  $\text{CH}_3\text{COOCH}_2\text{CH}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{OH}$   
**C**  $\text{CH}_2=\text{CH}_2 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$   
**D**  $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{CH}_3\text{CH}_2\text{OH} + 2\text{CO}_2$



## Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

| <b>A</b>                            | <b>B</b>                              | <b>C</b>                              | <b>D</b>                       |
|-------------------------------------|---------------------------------------|---------------------------------------|--------------------------------|
| <b>1, 2 and 3</b><br>are<br>correct | <b>1 and 2</b><br>only are<br>correct | <b>2 and 3</b><br>only are<br>correct | <b>1 only</b><br>is<br>correct |

No other combination of statements is used as a correct response.

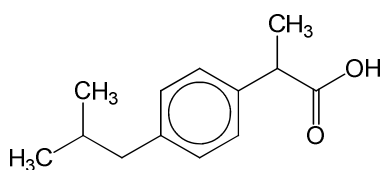
- 26** Complete combustion of a sample of hydrocarbon gave 0.132 g of carbon dioxide and 0.054 g of water. Which of the following formulae could correctly represent this hydrocarbon?

- 1**  $\text{CH}_2\text{CH}_2$   
**2**  $\text{CH}_3\text{CHCH}_2$   
**3**  $\text{CH}_3\text{CH}(\text{CH}_2\text{CH}_3)_2$

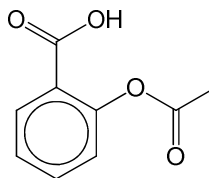
- 27** Why does aluminium chloride,  $\text{Al}_2\text{Cl}_6$ , sublime at a relatively low temperature of  $180^\circ\text{C}$ ?

- 1** Intermolecular forces between the  $\text{Al}_2\text{Cl}_6$  molecules are weak.  
**2** The dative bonds between Al and Cl atoms are weak.  
**3** The covalent bonds between Al and Cl atoms are weak.

- 28** Aspirin and ibuprofen are analgesic drugs used for pain, fever and inflammation. Their structures are shown below.



ibuprofen



aspirin

Which statements about **both** ibuprofen and aspirin are true?

- 1** They react with  $\text{LiAlH}_4$  to form alcohols.  
**2** They both decolourise hot acidified  $\text{KMnO}_4$ .  
**3** They undergo hydrolysis with hot, aqueous  $\text{NaOH}$ .

The responses **A** to **D** should be selected on the basis of

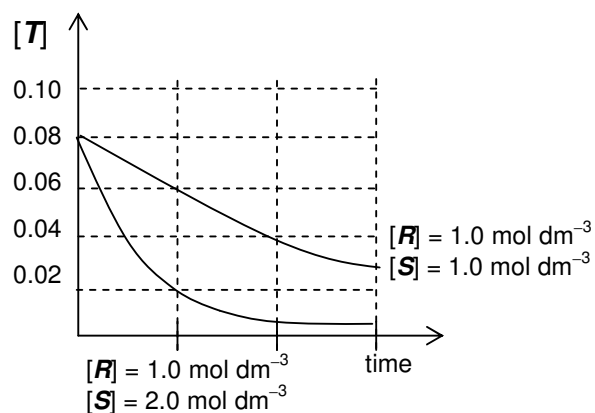
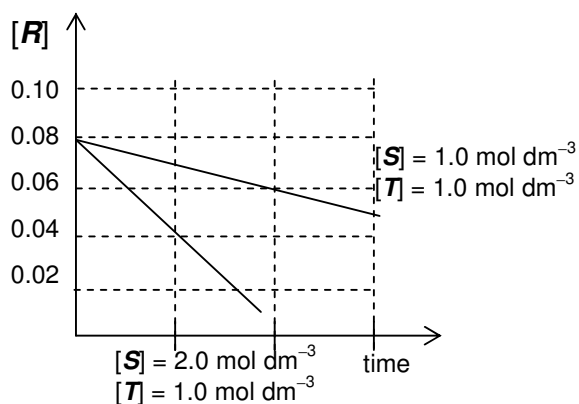
| <b>A</b>                            | <b>B</b>                              | <b>C</b>                              | <b>D</b>                       |
|-------------------------------------|---------------------------------------|---------------------------------------|--------------------------------|
| <b>1, 2 and 3</b><br>are<br>correct | <b>1 and 2</b><br>only are<br>correct | <b>2 and 3</b><br>only are<br>correct | <b>1 only</b><br>is<br>correct |

No other combination of statements is used as a correct response.

**29** Substances **R**, **S** and **T** react according to the following equation:



To find the rate equation for the above reaction, two sets of separate experiments were performed and the results are shown below.



What can be deduced from the information shown?

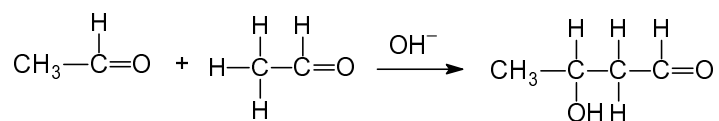
- 1** The order of reaction with respect to **T** is one.
- 2** **R** is not involved in the rate-determining step of the reaction.
- 3** The overall order of the reaction is two.

The responses **A** to **D** should be selected on the basis of

| <b>A</b>                            | <b>B</b>                              | <b>C</b>                              | <b>D</b>                       |
|-------------------------------------|---------------------------------------|---------------------------------------|--------------------------------|
| <b>1, 2 and 3</b><br>are<br>correct | <b>1 and 2</b><br>only are<br>correct | <b>2 and 3</b><br>only are<br>correct | <b>1 only</b><br>is<br>correct |

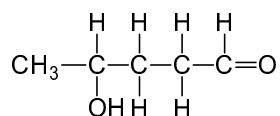
No other combination of statements is used as a correct response.

- 30** Under the influence of aqueous NaOH, two molecules of aldehyde may combine to form a 3-hydroxyaldehyde. This reaction is called aldol reaction. For example, the aldol reaction of ethanal is shown below.

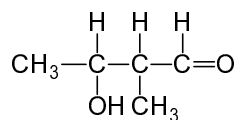


Which of the following molecules could be produced when a mixture of ethanal and propanal are warmed with aqueous NaOH?

**1**



**2**



**3**

